Date of Assignment: 15th June, 2026 **Submission Date:** 21 June, 2026

Course Title: Database Systems **Assessment :** Assignment 4

Instructions:

- Assignment should be hand written. Assignment should be submitted in soft form on the portal. Please upload a single PDF file.
- Also, submit the hard form assignment next day in the class. If you are unable to submit in class it would be consider as late.
- **This is group assignment of 4 students' assignment.**
- **No late submission at any cost.**

Question 1

Consider the following database schema and the functional dependencies. Normalize the given system till the highest normal form discuss in your class.

R = {Student_ID, L_Name, F_Name, st_DOB, st_phone, st_address, C_ID, C_Name, prof_id, pre_req, Grade, Prof_id, Prof_LName, Prof_FName, Prof_salary, Prof_phonenumber, Prof_emailid, Semester}

FD1: Student_ID → L_Name, F_Name , st_DOB, st_phone
FD2: st_phone → st_address
FD3: C_ID, Student_ID → grade
FD4: C_ID → C_Name , prof_id, pre_req
FD5: Prof_id → Prof_LName, Prof_FName, Prof_salary, Prof_phonenumber, Prof_emailid
FD6: Semester, C_id → Prof_id

Question 2

Identify the highest normal form in the following relations using the given FDs. Also determine the Candidate Key in each case.

a. R(A, B, C, D, E) $\{AC \rightarrow DF,\ D \rightarrow G,\ F \rightarrow BE\}$
b. R(U,V,W,X,Y,Z) $\{U \rightarrow W,\ UW \rightarrow VX,\ Y \rightarrow UX,\ Y \rightarrow Z,\ UW \rightarrow YZ\}$

Question 3

Consider the following table structure for storing information about books and authors:

BookID	Title	AuthorName	AuthorBirthdate	Publisher	PublisherAddress
1	Database Systems	C.J. Date	1941-12-23	Addison-Wesley	75 Arlington Street, Boston, MA
2	Database Systems	C.J. Date	1941-12-23	Addison-Wesley	75 Arlington Street, Boston, MA
3	Introduction to Algorithms	Thomas H. Cormen	1956-10-01	MIT Press	One Rogers Street, Cambridge, MA

1. Identify and explain the normal form the table currently adheres to.
2. Normalize the table to 1NF, 2NF, and 3NF.

Question 4

Consider the following unnormalized table:

OrderID	CustomerName	CustomerAddress	ProductID	ProductName	Quantity	UnitPrice
1	John Doe	123 Elm St	101	Laptop	1	1000
2	John Doe	123 Elm St	102	Mouse	2	50
3	Jane Smith	456 Oak St	103	Keyboard	1	75

Normalize this table to the Third Normal Form (3NF).

Question 5

Given the following table that stores student information:

StudentID	StudentName	CourseID	CourseName	InstructorName
1	Alice	101	Math	Dr. Smith
2	Bob	102	History	Dr. Johnson
1	Alice	102	History	Dr. Johnson

1. Identify any partial dependencies.
2. Identify any transitive dependencies.
3. Normalize the table to remove partial and transitive dependencies.